Evaluating AI Agents

Adding Structure to your Evaluations





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Talking Head

In this lesson:

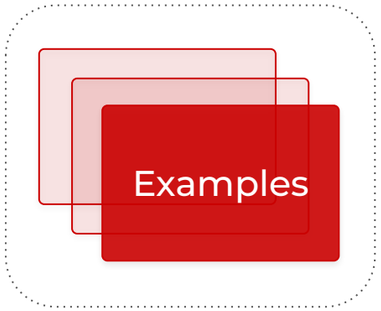
- Evaluation-Driven Development
- How to use experiments, datasets, and ground truth data to iterate on your agent

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Evaluation-Driven Development

Curate Dataset

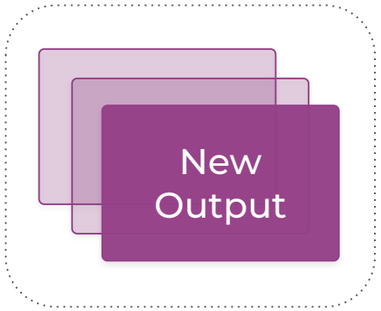


Track Changes as an Experiment

(Model, Prompt, Logic)

You're a helpful assistant. When user asks about return policy respond with {vars} ...

Evaluate the Experiment



Score

0.8

LLM APPS REQUIRE ITERATIVE PERFORMANCE IMPROVEMENTS

Curate Dataset

Construct a dataset of **test cases**:

- 25+ examples of inputs to your agent
- Should be **comprehensive**
- Can be sourced from live runs of your agent, or constructed beforehand
- Attached **expected outputs** to these test cases
  - *Note: LLM-as-a-Judge evals can be used without an expected output to compare against.*

Track Agent Changes

Test proposed changes to your agent:

- Prompt iterations
- Tool definitions
- Router logic
- Skill structure
- Model changes

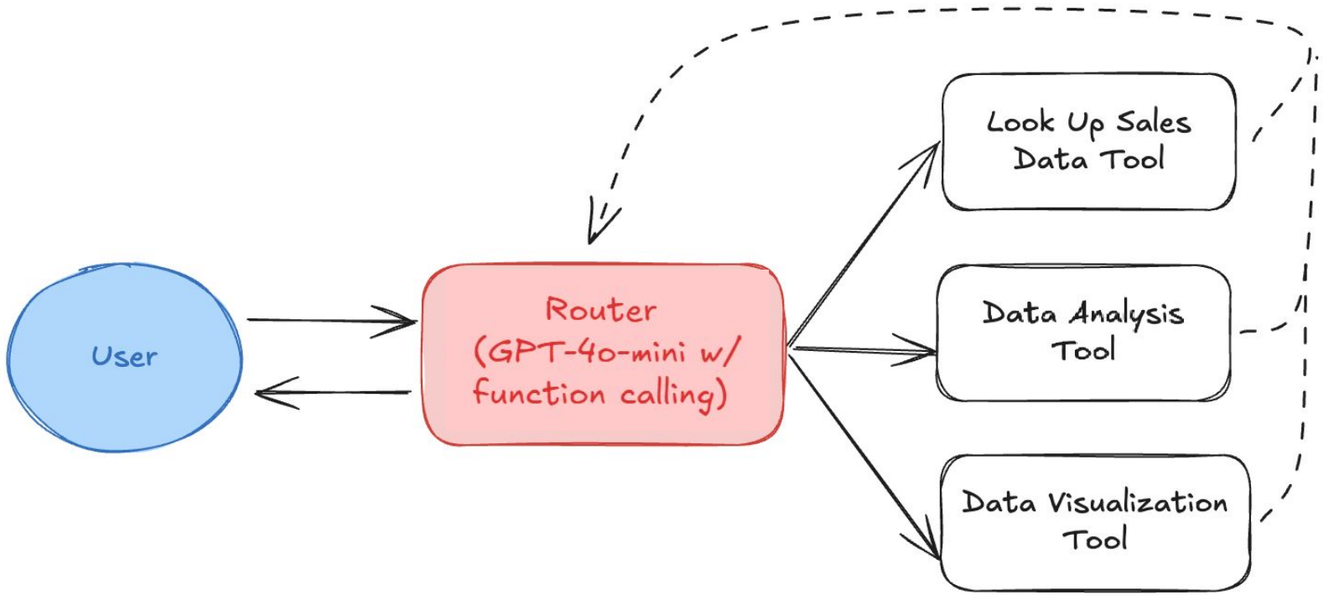
Experiments are a structure used to run each of your curated dataset test cases through your agent, and record the results

Evaluate your Experiment

Use your same evaluators from previous lessons:

- Code-based evals
  - Compare against ground truth
  - Generated code is runnable
  - Convergence
- LLM-as-a-Judge evals
  - Function calling
  - Analysis clarity
  - Entity correctness

Applying Evaluation Driven Development to your Agent



Applying Evaluation Driven Development to your Agent

Each aspect of your agent can be built into an experiment:

Component of your agent	**Example test case**	**Experiment**	**Evaluations**
Router	{ 'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'database_lookup'}	Testing different tool description strings	1. Code-based comparison against ground truth 2. Function calling using LLM-as-a-Judge

Applying Evaluation Driven Development to your Agent

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Database lookup tool	{'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'SELECT Store_Number, SUM(Qty_Sold) AS Total_Qty_Sold,....'}	?	?

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Database lookup tool	{ 'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'SELECT Store_Number, SUM(Qty_Sold) AS Total_Qty_Sold,....'}	Testing different SQL generation prompts	1. Code-based comparison against ground truth
Database analysis tool	{ 'input': '{'role': 'user', 'content': 'Which stores had the best sales performance in 2021'}, {'role': 'tool content': 'Store_Number Total_Qty_Sold Total_Sale_Value 0 2970 5867.0 84454.329826....'}'}	?	?

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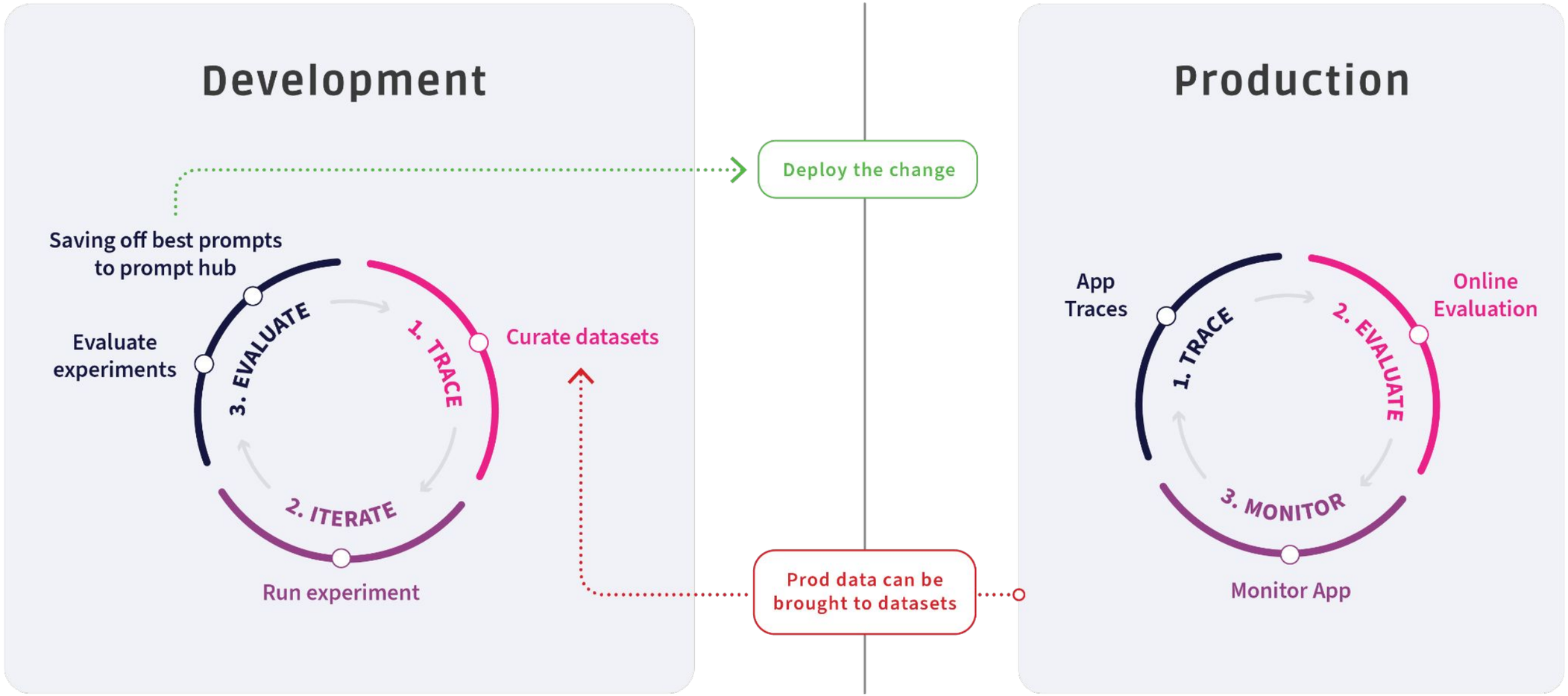
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Comparing your Results



name	description	created at	code_is_runnable	evaluate_clarity	evaluate_entity_correctness	evaluate_sql_result	function_calling_eval	run count	avg latency	error rate
#2 Overall Experiment	Evaluating the overall experiment	1/29/2025, 04:25 PM	0.86	0.86	0.86	0.43	0.71	7	8.14s	14.29%
#1 Overall Experiment	Evaluating the overall experiment	1/29/2025, 04:20 PM	0.86	0.86	0.86	0.57	0.79	7	6.86s	14.29%

Building across Development and Production



What you've learned:

1. The purpose of Evaluation-Driven Development and how it works
2. How to structure experiments around your evals